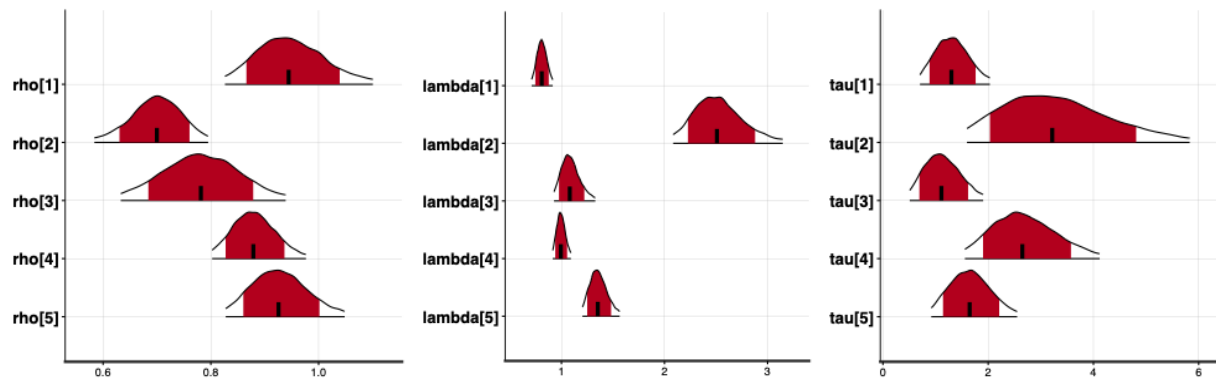


HW5

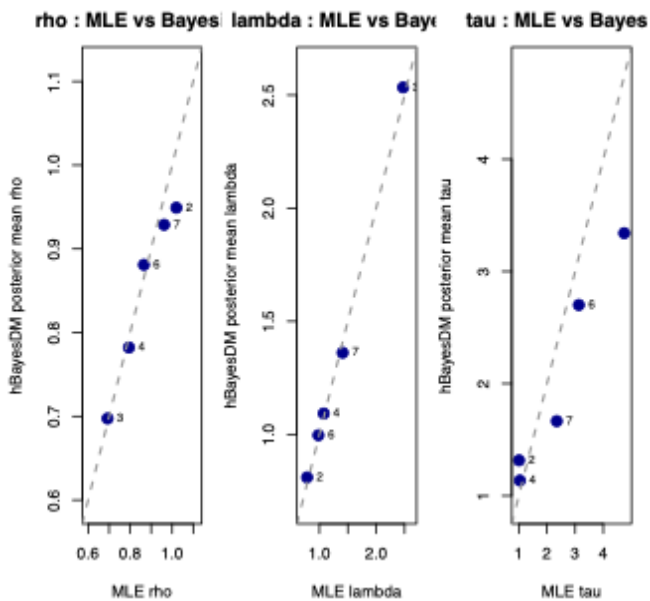
Dongwook Hwang

Q1_(1)



rho[1] = subj_2, rho[2] = subj_3, rho[3] = subj_4, rho[4] = subj_6, rho[5] = subj_7

Q1_(2)



x-axis: MLE estimation

y-axis: hBayesDM posterior mean

Q1_(3)

The MLE estimates from HW2 and the hBayesDM posterior means showed broadly similar values, but with one clear difference: subj_3's λ shifted from 2.47 (MLE) to about 2.2 (Bayesian), while the other four subjects' estimates remained nearly unchanged across both approaches. This pattern reflects shrinkage, the key feature of hierarchical Bayesian analysis.

The degree of shrinkage also depended on the parameter. For ρ , all five MLE estimates fell within a narrow range (0.69–1.02), so the two approaches agreed closely. For τ , the wider spread (1.01–4.74) led to a more diffuse prior, preserving individual estimates. λ showed the strongest shrinkage because of the clear contrast between subj_3 and the rest.

Q2

Q2

$$(a) BIC_1 (\text{re-prospect}) = 580.434$$

$$BIC_2 (\text{re-noA}) = 651.866$$

$$BF_{12} = \exp((BIC_2 - BIC_1)/2)$$

$$= \exp(71.432/2) = \exp(35.716)$$

$$\therefore BF_{12} \approx 3.25 \times 10^{15}$$

$$(b) \text{Jeffery's evidence: } BF > 100 \rightarrow \text{decisive}$$

$$BF_{12} \approx 3.25 \times 10^{15} > 100$$

$$\therefore \text{Decisive evidence for Model 1 (re-prospect)}$$

$$(c) BF_{12} = P(D|M_1) / P(D|M_2)$$

$$\text{posterior odds} = P(M_1|D) / P(M_2|D) = \text{Bayes factor} \times \text{prior odds}$$

$$\therefore \frac{P(M_1|D)}{P(M_2|D)} = BF_{12} \times \frac{0.6}{0.4} = 3.25 \times 10^{15} \times 1.5$$

$$\text{posterior odds} \approx 4.88 \times 10^{15}$$

$$(d) P(M_1|D) = \frac{\text{posterior odds}}{1 + \text{posterior odds}} = \frac{4.88 \times 10^{15}}{1 + 4.88 \times 10^{15}} \approx 1.0$$

$$\therefore \text{Model 1 is superior to Model 2}$$

$$(e) BF_{\bar{z}\bar{j}} = \exp((BIC_{\bar{z}} - BIC_{\bar{j}})/2)$$

$$\bar{z} = \text{POW2} \quad \bar{j} = \text{POW1} \quad BF_{\bar{z}\bar{j}} = 1$$

$$= \text{POW2} \quad \approx 2.83$$

$$= \text{EXP1} \quad \approx 5.3 \times 10^{10}$$

$$= \text{EXP2} \quad \approx 1.1 \times 10^3$$

$$= \text{EXPW} \quad \approx 8.02$$

$$= \text{HYP1} \quad \approx 46.2$$

$$= \text{HYP2} \quad \approx 23.9$$

Q3_(1), (2)

	mean	se_mean	sd	2.5%	25%		50%	75%	97.5%	n_eff	Rhat
rho[1]	1.03	0.00	0.10	0.87	0.96	rho[1]	1.02	1.09	1.26	6433	1
rho[2]	0.68	0.00	0.05	0.57	0.65	rho[2]	0.69	0.71	0.76	9217	1
rho[3]	0.79	0.00	0.11	0.60	0.72	rho[3]	0.79	0.86	1.03	9391	1
rho[4]	0.87	0.00	0.04	0.79	0.84	rho[4]	0.87	0.89	0.96	10218	1
rho[5]	0.96	0.00	0.07	0.84	0.91	rho[5]	0.95	1.00	1.12	7496	1
lambda[1]	0.79	0.00	0.06	0.67	0.75	lambda[1]	0.79	0.82	0.90	10691	1
lambda[2]	2.53	0.00	0.27	2.11	2.36	lambda[2]	2.50	2.66	3.17	5455	1
lambda[3]	1.13	0.00	0.15	0.93	1.04	lambda[3]	1.10	1.18	1.46	4885	1
lambda[4]	0.99	0.00	0.04	0.91	0.96	lambda[4]	0.99	1.01	1.07	14007	1
lambda[5]	1.38	0.00	0.11	1.21	1.31	lambda[5]	1.37	1.44	1.62	8626	1
tau[1]	1.01	0.00	0.36	0.40	0.75	tau[1]	0.98	1.24	1.80	8374	1
tau[2]	4.56	0.02	1.47	1.96	3.53	tau[2]	4.45	5.50	7.73	8019	1
tau[3]	1.00	0.00	0.41	0.30	0.71	tau[3]	0.97	1.27	1.88	9436	1
tau[4]	3.12	0.01	0.73	1.80	2.61	tau[4]	3.08	3.59	4.65	10750	1
tau[5]	1.54	0.01	0.48	0.70	1.19	tau[5]	1.51	1.84	2.54	7762	1
lp__	-268.23	0.05	3.12	-275.33	-270.06	lp__	-267.85	-265.95	-263.29	3548	1

rho[1] = subj_2, rho[2] = subj_3, rho[3] = subj_4, rho[4] = subj_6, rho[5] = subj_7